

Monday, May 4th

Sample space: The set of all possible outcomes of a random experiment.

Examples:

① we toss a coin once: $S = \{H, T\}$

② we toss a coin twice:

$$S = \{HH, TT, HT, TH\}$$

③ we toss a coin until we observe a head.

$$S = \{H, TH, TTH, TTTTH, \dots\}$$

Events: If a sample space is countable (discrete) any subset of S is called an event.

Simple events: They consist of one outcome (sample point)

Compound events: They consist of multiple outcomes (sample points)

Example:

toss a coin twice $S = \{HH, HT, TH, TT\}$

$A =$ observe two heads. $= \{HH\} \rightarrow$ Simple event

$B =$ observe at least one heads

$B = \{HT, TH, HH\} \Rightarrow$ compound event

$$1) P(\emptyset) = 0 \quad \emptyset \cap S = \emptyset$$

$$P(S) = P(S \cup \emptyset) = P(S) + P(\emptyset) \Rightarrow P(\emptyset) = 0$$

Alternatively.

$$\emptyset = \emptyset \cup \emptyset \cup \emptyset \dots \quad , \quad \emptyset \cap \emptyset = \emptyset$$

$$P(\emptyset) = P(\emptyset \cup \emptyset \cup \emptyset \dots)$$

$$= P(\emptyset) + P(\emptyset) + \dots \quad \left. \begin{array}{l} \text{Axiom 3} \end{array} \right\}$$

$$\Rightarrow P(\emptyset) = 0 \quad \leftarrow \text{Axiom 1}$$

2) For finite sequence of mutually exclusive events

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

$$A_{n+1} = A_{n+2} = \dots = \emptyset$$

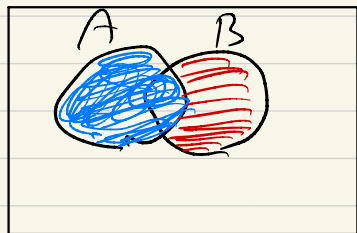
$$P\left(\bigcup_{i=1}^n A_i\right) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(A_i)$$

3) let A and B be arbitrary events (Additive law of probability)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

least one events occur

$$\begin{aligned} A \cup B &= A \cup (B - A) \\ &= A \cup (B \cap A^c) \end{aligned}$$

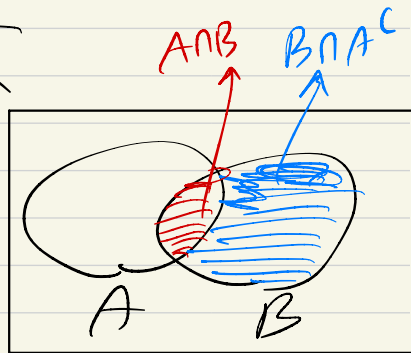


$$P(A \cup B) = P(A) + P(B \cap A^c)$$

$$B = (B \cap A^c) \cup (A \cap B)$$

$$P(B) = P(B \cap A^c) + P(A \cap B)$$

$$P(B) - P(A \cap B) = P(B \cap A^c)$$



$$\rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$4) \text{ If } A \subset B \Rightarrow P(A) \leq P(B)$$

$$B = A \cup (B \cap A^c)$$



$$P(B) = P(A) + P(B \cap A^c)$$

$\downarrow$ $\downarrow$

≥ 0 ≥ 0

by Axiom 1

Hence

$$P(B) \geq P(A)$$

5) Show that $P(A) \leq 1$

We know that $A \subset S$, hence by the previous result,

$$P(A) \leq P(S) = 1$$

└──────────→ Axiom 2.

6) $P(A^c) = 1 - P(A)$

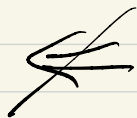
$$S = A \cup A^c \quad A \cap A^c = \emptyset$$

$$\begin{aligned} 1 = P(S) &= P(A \cup A^c) \\ &= P(A) + P(A^c) \end{aligned}$$

Hence, $P(A^c) = 1 - P(A)$

Note

$$A = \phi \Rightarrow P(A) = 0$$



$$A = S \Rightarrow P(A) = 1$$



Definition (Equilikely Case)

Suppose $S = \{x_1, x_2, \dots, x_m\}$ be a finite sample space and $p_i = \frac{1}{m} = P(x_i)$, $i = 1, \dots, m$ and for all subsets of S , say A ,

$$P(A) = \sum_{x_i \in A} \frac{1}{m} = \frac{\#(A)}{m}$$

where $\#(A)$ is the number of sample points (elements) in A . Then P is a probability on S and it's called an equilikely case.

Example.

Suppose we toss a fair coin twice.

$$S = \{HH, TH, HT, TT\}$$

Find the probability that we observe two heads

$$A = \{HH\} \quad P(A) = \frac{1}{4}$$

Find the probability that we observe at least one heads

$$B = \{HT, TH, HH\} \quad P(B) = \frac{3}{4}$$

Suppose we have n objects, we want to sample r objects.

without replacement

- order is important (Permutation)
- order is not important (Combination)

with replacement

- order is important
 $n \times n \times \dots \times n = n^r$
- order is not important

$$x_1 + x_2 + \dots + x_n = r$$

$$\binom{n+r-1}{n-1}$$

$\hookrightarrow$ non-negative integer solutions

$$P_r^n = n \overset{1^{st}}{\times} (n-1) \overset{2^{nd}}{\times} (n-2) \times \dots \times (n-r+1) \overset{r^{th}}{\uparrow}$$

$$= \frac{n!}{(n-r)!}$$

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$$

$$0! = 1, 1! = 1, 2! = 2 \times 1 = 2$$

$$C_r^n = \binom{n}{r} = \frac{n(n-1) \times \dots \times (n-r+1)}{r \times (r-1) \times \dots \times 2 \times 1}$$

$$= \frac{n!}{r! (n-r)!}$$

Examples,

Choose two letters from A, B, C

① without replacement, order is important
(permutation)

$\{A, B\}, \{B, A\}$

$$P_2^3 = \frac{3!}{(3-2)!} = 3! = 6$$

$\{A, C\}, \{C, A\}$

$\{B, C\}, \{C, B\}$

② without replacement, order is not important

$\{A, B\}, \{A, C\}, \{B, C\}$

$$C_2^3 = \frac{3!}{2!(3-2)!} = 3$$

note: $\binom{n}{2} = \frac{n(n-1)}{2}$

③ with replacement, order is important.

$\{A, B\}, \{B, A\}, \{A, A\}$

$\{A, C\}, \{C, A\}, \{B, B\}$

$\{B, C\}, \{C, B\}, \{C, C\}$

④ with replacement, order is not important

$\{A, B\}, \{A, A\}$

$\{A, C\}, \{B, B\}$

$\{B, C\}, \{C, C\}$

$$\binom{3+2-1}{3-1} = \binom{4}{2} = 6$$

$$x_1 + x_2 + x_3 = 2$$

$$(1, 1, 0)$$

$$(1, 0, 1)$$

$$(0, 1, 1)$$

$$(0, 0, 2)$$

$$(0, 2, 0)$$

$$(2, 0, 0)$$

binomial coefficient. $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

$$(x+y)^2 = x^2 + 2xy + y^2$$

$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $1 = \binom{2}{0} \quad \quad \quad 2 = \binom{2}{1} \quad \quad \quad 1 = \binom{2}{2}$

$$(x+y)^3 = x^3 + 3xy^2 + 3x^2y + y^3$$

$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $1 = \binom{3}{0} \quad \quad \quad 3 = \binom{3}{1} \quad \quad \quad 3 = \binom{3}{2} \quad \quad \quad 1 = \binom{3}{3}$

Example 1.3.3

If there are n people in a room. Find the probability that at least two people have the same birthday.

$$P[A] = 1 - P(A^c) = 1 - P\left(\begin{array}{l} \text{All people} \\ \text{have different} \\ \text{birthday.} \end{array}\right) = 1 - \frac{P_n^{365}}{365^n}$$

$$= 1 - \frac{365 \times 364 \times \dots \times (365 - n + 1)}{365^n}$$

Example.

Suppose we have 4 stats books,

3 math books, 2 physics books

We randomly place them on a shelf. Find a probability that books from the same are placed next to each other

$$P(A) = \frac{3! (4! \times 3! \times 2!)}{9!}$$

Example: Find the number of 4-letter words using the letters in Omid, each letter is used only once. $4! = 4 \times 3 \times 2 \times 1 = \binom{4}{1, 1, 1, 1}$

what if it's Adam? $\frac{4!}{2!} = \binom{4}{2, 1, 1}$

make 11-letter words using Mississauga.

what about mississauga.

$$\binom{11}{4, 2, 2, 1, 1, 1} = \frac{11!}{4! 2! 2!}$$

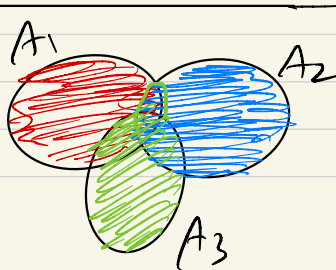
Inclusion-exclusion theorem

$$P\left(\bigcup_{i=1}^k A_i\right) = \sum_{i=1}^k P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \dots + (-1)^{k-1} P(A_1 \cap \dots \cap A_k)$$

proof: Exercise

Hint: Induction

$k=3$



theorem: (Continuity of probability Function)

Let $\{C_n\}$ be an increasing sequence of events

$$\text{then } \lim_{n \rightarrow \infty} P(C_n) = P(\lim_{n \rightarrow \infty} C_n) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$$

proof

from the previous section $\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n$

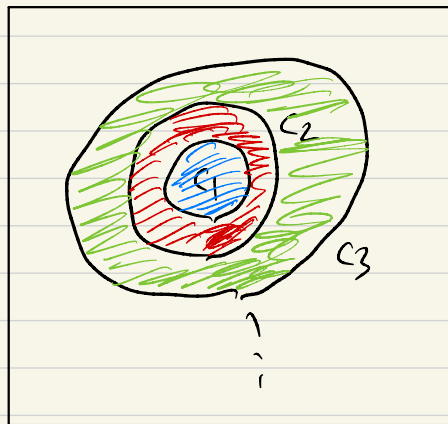
$$\text{Hence } P(\lim_{n \rightarrow \infty} C_n) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$$

$$R_1 = C_1 \quad \checkmark$$

$$R_2 = C_2 - C_1 = C_2 \cap C_1^c$$

$$R_3 = C_3 - C_2 = C_3 \cap C_2^c$$

$$\vdots$$
$$\bigcup_{n=1}^{\infty} R_n = \bigcup_{n=1}^{\infty} C_n$$



$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\bigcup_{n=1}^{\infty} R_n\right) = \sum_{n=1}^{\infty} P(R_n) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(R_i)$$

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(R_i)$$

$$= \lim_{n \rightarrow \infty} \left[P(R_1) + \sum_{i=2}^n P(R_i) \right]$$

$$= \lim_{n \rightarrow \infty} \left[\cancel{P(C_1)} + \sum_{i=2}^n \underbrace{P(C_i) - P(C_{i-1})}_{\substack{P(C_2) - P(C_1) \\ + P(C_3) - P(C_2) \\ \vdots \\ P(C_n) - P(C_{n-1})}} \right]$$

$$= \lim_{n \rightarrow \infty} P(C_n)$$

$$C = A - B = A \cap B^c$$

$$P(C) = P(A) - P(B) \quad X$$

$$\text{if } B \subset A$$

$$P(C) = P(A) - P(B)$$

Here is why

$$P(C) = P(A \cap B^c) = P(A) - P(A \cap B)$$

$$\text{if } B \subset A \Rightarrow A \cap B = B$$

$$P(C) = P(A) - P(B)$$

Theorem: Boole's Inequality

Let $\{A_n\}$ be a sequence of arbitrary events

then
$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n)$$

$$R_1 = A_1$$

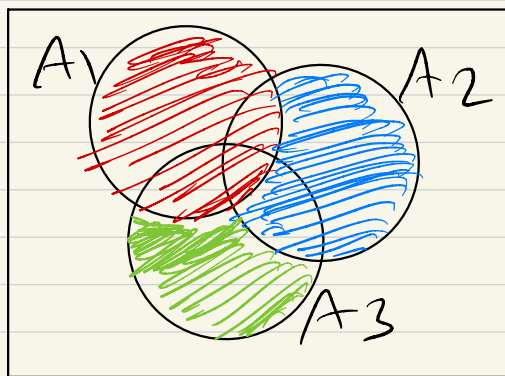
$$\begin{aligned} R_2 &= A_2 - A_1 \\ &= A_2 \cap A_1^c \end{aligned}$$

$$R_3 = A_3 - A_1 \cup A_2$$

!

$$R_n = A_n - \bigcup_{i=1}^{n-1} A_i$$

1
2



$$\bigcup_{n=1}^{\infty} R_n = \bigcup_{n=1}^{\infty} A_n$$

$$P(\bigcup_{n=1}^{\infty} A_n) = P(\bigcup_{n=1}^{\infty} R_n)$$

$$\text{Axiom 3} \quad = \sum_{n=1}^{\infty} P(R_n) \\ \leq \sum_{n=1}^{\infty} P(A_n)$$

$$R_i \subset A_i, \quad i = 2, 3, \dots$$

$$R_1 = A_1 \quad \text{Hence, } P(R_i) \leq P(A_i)$$

Recall that if $A \subset B \Rightarrow P(A) \leq P(B)$

Bonferroni's Theorem.

Recall that if A is an event $P(A) \leq 1$

Also, if C_1 and C_2 are two events

$$P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$$

Since, $P(C_1 \cup C_2) \leq 1$, we have

$$P(C_1) + P(C_2) - P(C_1 \cap C_2) \leq 1$$

Hence, $P(C_1 \cap C_2) \geq P(C_1) + P(C_2) - 1$

which provides a lower bound for $P(C_1 \cap C_2)$

Recall that an upper bound for $P(C_1 \cap C_2)$ is $\min \{P(C_1), P(C_2)\}$